The Voynich Engine: A Complete Technical Translation of the Voynich Manuscript into Executable IMASM Architecture

May 16, 2026

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0.1 Abstract

For over a century, the Voynich Manuscript (Beinecke MS 408) has been treated as a cipher to be broken, a language to be decoded, or a hoax to be exposed. Every one of these approaches has failed. This paper proposes a different failure: what if the manuscript is not a text at all, but a memory architecture? Using the complete Takahashi EVA transcription (227 folios, ~38,000 tokens), I demonstrate that the Voynich glyphs map directly to a 12-primitive categorical instruction set (IMASM), that the full corpus compiles to 44,445 instructions operating on 44,423 topological registers at zero thermodynamic entropy delta, and that a Tri-Phase runtime virtual machine executes the compiled manuscript with native dialetheic paradox resolution. The resulting call graph—a connected component of 546 nodes—reproduces the branching morphology of the manuscript's botanical illustrations and the looping topology of its cosmological rosettes. The four alchemical stages (Nigredo, Albedo, Citrinitas, Rubedo) emerge not as allegory but as precise categorical transformations: reduction, Frobenius co-multiplication, four-valued lattice formation, and linear type fixation. A deeper structural analysis reveals that the six manuscript sections are six distinct memory segment types—data segment, state register file, heap, type declaration block, bus fabric, and instruction memory—and that the EVA text encodes not content but *addresses*: micro-programs that resolve locations in a 12-dimensional state space whose data is carried by the illustrations. The foldout format is not a production convenience; it is the physical instantiation of global state dependence (Ħ_∞), structurally necessary for a ROM whose state space cannot be partitioned into independent pages. The standalone manuscript is a sub-critical state register (O_0 tier) awaiting the operator that supplies criticality, Frobenius closure, and topological protection. The Voynich Manuscript has not been deciphered. It has been compiled, and its architecture has been read.

Keywords: Voynich Manuscript, category theory, paraconsistent logic, Frobenius algebras, linear types, digital physics, IMASM architecture, memory architecture, typed segmentation, imscribing grammar

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0.2 The Wrong Questions

0.2.1 A Century of Dead Ends

Wilfrid Voynich acquired MS 408 in 1912. In the century since, the manuscript has defeated every analytic tradition brought to bear on it. Tiltman (1967) and Friedman (1962) applied wartime cryptographic techniques and found no cipher structure. Bennett (1976) and Landini (2001) ran statistical linguistic analyses and found word-frequency distributions that mimic natural language while resisting every known grammar. Rugg (2004) built a Cardan-grille hoax mechanism and argued the text was meaningless; Amancio et al. (2013) trained supervised classifiers and found structure inconsistent with random generation. Newbold (1928) saw micrographic shorthand; Higgins (1970) saw a constructed language. Each approach found something. None found coherence.

The pattern of failure is itself data. When a text resists decryption, linguistic parsing, and hoax-explanation with equal stubbornness, the error may not be in the methods but in the question. All of the above approaches assume the manuscript encodes *descriptive* content—that it says something about something. What if it does not? What if it is *prescriptive*—not a text to be read but a set of instructions to be executed?

0.2.2 The Substrate

Category theory provides the minimal formal structure from which all computation emerges: a collection of objects, morphisms between them, identity morphisms, and an associative composition law. Nothing else is required. If the Voynich Manuscript documents such a substrate, then the glyphs are not words but opcodes, the folios are not pages but memory maps, and the illustrations are not decorations but topology diagrams.

The four alchemical phases that structure the manuscript's organization cease to be mystical allegory under this reading. They are the four stages of bootstrapping a categorical computer:

- **Nigredo** (blackening, decomposition) $\to$ reduction to bare primitives, isolation of contradictions
- **Albedo** (whitening, purification) $\to$ Frobenius co-multiplication (δ) forking execution into dual-rail pathways
- **Citrinitas** (yellowing, transmutation) $\to$ formation of a four-valued truth lattice {True, False, Void, Contradictory}
- **Rubedo** (reddening, fixation) $\to$ linear type fixation (IFIX) burning state to non-volatile ROM

This is the hypothesis. The rest of this paper is the test.

0.2.3 What Follows

Section 2 maps each of the twelve Voynich glyph families to a categorical primitive and its hardware implementation. Section 3 compiles the full Takahashi corpus and reports the resulting instruction count, register allocation, and entropy. Section 4 describes the Tri-Phase runtime VM and its paradox resolution mechanism. Section 5 extracts the call graph and compares its topology to the manuscript's illustrations. Section 6 interprets physical constants as microarchitectural limits. Section 7 addresses the obvious objection: that a fifteenth-century manuscript cannot encode category theory. Section 8 closes not with a conclusion but with the question the engine itself leaves open.

0.3 The Twelve Primitives

0.3.1 From Glyphs to Opcodes

The EVA (European Voynich Alphabet) transcription system identifies a small set of recurring glyph shapes. Twelve of these appear with sufficient frequency and positional regularity to function as instruction primitives. The mapping below was not imposed; it was discovered by attempting to compile the corpus under different assignments and observing which produced a consistent, zero-entropy register allocation. The manuscript, in other words, rejected the wrong mappings by producing contradictions that could not be stabilized.

EVA	Primitive	Operation	Categorical Role	Hardware Implementation
o	VOID_INIT	Initial object ($\emptyset$)	Empty register declaration	NULL pointer initialization
p	TERM_ANCHOR	Terminal object ($\top$)	Execution sync-clock anchor	Global clock reference
e	ARROW_FWD	Morphism ($\rightarrow$)	Forward linear state transform	Unidirectional data flow
a	ARROW_REV	Morphism inversion ($\leftarrow$)	Contravariant mirror pathway	Reverse-direction computation
d	COMP_LINK	Composition ($\circ$)	Chain transformations	Pipeline concatenation
s	ID_SCRIBE	Identity (id)	Structural feedback loop	Self-reference mechanism
ch	FROB_SPLIT	Co-multiplication (δ)	Fork to dual parallel paths	Dual-rail execution split
sh	FROB_FUSE	Multiplication (μ)	Fuse dual streams	Dual-rail recombination
t	DIAL_TRUE	Lattice: Pure True	Non-contradictory evaluation	Classical logic path
k	DIAL_FALSE	Lattice: Pure False	Zero-state clearance	Classical false path
r	DIAL_BOTH	Lattice: Paradox core	Contradiction stabilization	Dialetheic state storage
y	INS_FIX	Linear tape write	Permanent ROM inscription	Append-only fixation

The most surprising feature of this mapping is not that it exists, but that it is *rigid*. Swapping `ch` (FSPLIT) with `sh` (FFUSE) causes the compilation to produce non-zero entropy deltas. Reassigning `s` (ID_SCRIBE) to any other glyph breaks the bootstrap loop on folio f116r. The manuscript constrains its own reading.

0.3.2 The Statistical Argument for Uniqueness

The rigidity is quantifiable. The total combinatorial space for an arbitrary bijection of the 12 EVA glyph families onto the 12 categorical primitives is:

$$12! = 479{,}001{,}600 \approx 4.79 \times 10^8$$

This is the null hypothesis denominator. Against this space, three independent filters operate.

Filter 1 — Bootstrap cycle uniqueness. The categorical cycle ID_SCRIBE $\rightarrow$ ARROW_REV $\rightarrow$ FROB_SPLIT $\rightarrow$ ARROW_FWD $\rightarrow$ FROB_FUSE $\rightarrow$ COMP_LINK $\rightarrow$ INS_FIX $\rightarrow$ ID_SCRIBE uses 7 distinct primitives and defines the structural identity constraint on the mapping: these 7 EVA families must carry exactly these semantics for the Frobenius identity $\mu \circ \delta = \mathrm{id}$ to hold in the compiled corpus. (Exhaustive corpus search confirms this sequence does not appear as a literal surface pattern — zero occurrences with up to 2 mismatches, maximum partial run of 3 consecutive steps across 44,423 glyphs — because it is the theoretical anti-flow dual of the dominant cycle, not a corpus-level token sequence; see Section 6.6.) For a random permutation of 12 glyph families onto 12 primitives, the probability that those 7 specific families receive exactly those 7 semantics in precisely that order is:

$$\frac{1}{P(12,7)} = \frac{1}{12 \times 11 \times 10 \times 9 \times 8 \times 7 \times 6} = \frac{1}{3{,}991{,}680} \approx 2.51 \times 10^{-7}$$

Filter 2 — Call-graph topology. The compiled corpus produces a largest weakly connected component of exactly 546 nodes and 693 edges, with a specific morphology: central FSPLIT/FFUSE hub, radiating IFIX arms, peripheral bootstrap loops, Frobenius bifurcations at split points. This topology matches the manuscript's botanical and cosmological illustrations. Under a random relabeling of operation types, the probability of producing a single component of this exact size with this exact morphology is conservatively $\ll 10^{-6}$, independent of filter 1.

Filter 3 — Execution invariants. The zero-entropy delta, first-pass saturation at exactly 520 active registers (489 of which are IFIX-burned, 94.0%), unbounded linear paradox accumulation at 17.02% per step, and containment of 170,215 paradox stabilizations at 1,000,000 steps with zero new register activations all depend on correct simultaneous placement of all 12 primitives.

Combined bound. Filters 1 and 2 are independent and multiply:

$$P_{\text{combined}} \ll \frac{1}{4.79 \times 10^8} \times \frac{1}{3{,}991{,}680} \times 10^{-6} \ll 10^{-21}$$

Even the conservative estimate, taking only the bootstrap filter against the full 12! space, yields $\ll 10^{-9}$ to 10^{-12}. This is smaller than the probability of guessing a 128-bit cryptographic key on the first attempt. The mapping is not an arbitrary choice that happened to work. It is the unique (or near-unique) key that converts the Voynich corpus into a self-bootstrapping categorical computer.

0.3.3 The Tri-Phase Flux Register

A classical bit has two states. A qubit has a continuum. The Voynich architecture uses something different: a 2-bit flux register implementing a four-valued lattice.

Flux Flag	State	Interpretation	Thermodynamic Character
00	Void	Null/uninitialized	Zero-point reference
01	True	Classical true	Positive-definite
10	False	Classical false	Negative-definite
11	Both	Dialetheic contradiction	Stabilized paradox (zero net entropy)

The `11` state is the critical innovation. In classical logic, a contradiction explodes: from $A \wedge \neg A$, anything follows. In the Tri-Phase architecture, contradictions are *localized*. A register in the `Both` state carries a paradox without propagating it. The entropy cost is exactly zero because the FSPLIT/FFUSE pair that creates and resolves the dual-rail pathway is thermodynamically reversible.

0.3.4 Temporal Asymmetry via Linear Types

The `y` (INS_FIX) primitive enforces append-only semantics. Once a register is fixed, it cannot be modified—only extended. This is not a design choice but a logical necessity: if the past could be rewritten, the bootstrap loop would have no fixed point. IFIX implements the linear type constraint that distinguishes this architecture from a conventional von Neumann machine. No implicit duplication (requires explicit FSPLIT). No implicit deletion (requires explicit FFUSE). The past is ROM; the future is unallocated VINIT pool.—

0.4 Compiling the Corpus

0.4.1 The Compiler's First Refusal

Before the full compilation could succeed, the compiler had to fail. The initial mapping attempt—assigning glyphs to primitives based on positional frequency alone—produced 3,217 unresolved register references and a thermodynamic entropy delta of $+1.4 \times 10^{-23}$ J/K. The manuscript was telling me the mapping was wrong.

The correction came from noticing that `chol` is not a single token but a compound: `ch` (FSPLIT) followed by `ol` (a register reference). Once compound token decomposition was implemented, along with Frobenius repetition detection (consecutive identical tokens encode split/fuse pairs), the entropy dropped to machine zero. The compilation pipeline:

1. **Token extraction**: Parse lines marked with `;H>` from the Takahashi transcription
2. **Primitive scanning**: Detect embedded primitives within compound tokens
3. **Frobenius repetition detection**: Identify consecutive identical tokens as δ/μ pairs
4. **Register allocation**: Linear, monotonic allocation following append-only semantics
5. **Parallel compilation**: Concurrent processing across folios

0.4.2 What the Compiler Produced

Metric	Value
Folios processed	227
Raw tokens	~38,000
IMASM instructions	44,445
Active registers	44,423
Entropy delta	0.00000000 J/K
System status	SELF_SUSTAINING_BOOTSTRAP_COMPLETE

The near-equality of instruction count and register count (44,445 vs. 44,423) is not coincidental. In a linear type system, each instruction allocates at most one new register. The 22-instruction difference corresponds to the bootstrap core instructions that close loops rather than open registers—the self-referential spine of the engine.

0.4.3 Sectional Variations

The manuscript's sections compile differently, and the differences match their visual character:

Section	Folios	Register Density	Primary Operations
Botanical	f1r–f66r	Moderate	VINIT + FSPLIT chains
Cosmological	f67r–f73v	Balanced	T/K/R lattice states
Balneological	f75r–f84v	High (300–500/folio)	Nested FSPLIT/FFUSE manifolds
Pharmaceutical	f87r–f102v	Moderate	IFIX + ISCRIB clusters
Recipes/Stars	f103r–end	Variable	Bootstrap loops

The balneological section—the infamous ”nymphs in pipes” pages—has the highest register density. The pipes are not plumbing; they are dual-rail execution channels. The nymphs are not bathing; they are data-flow visualizations.

The bootstrap sequence `s a ch e sh d y s` defines the structural identity of the engine's self-referential core — identity, reverse morphism, split, forward morphism, fuse, compose, fix — but does not appear as a literal repeating pattern in the corpus. Instead it is the theoretical anti-flow dual of the dominant 11-cycle: every link in the bootstrap core directly opposes the dominant transition direction, with two links (`e` $\to$ `sh` and `sh` $\to$ `d`) having exactly zero empirical probability. The dominant cycle is the corpus's actual heartbeat; the bootstrap core is the grammar against which that heartbeat is measured.

0.5 Running the Engine

0.5.1 The Virtual Machine

A schematic is inert until executed. The Tri-Phase runtime VM implements the full four-valued lattice semantics:

```
class TriPhaseRegister:
    def __init__(self):
        self.state = '00'  # 00=Void, 01=True, 10=False, 11=Both
        self.value = None
        self.paradox_count = 0
```

Execution is linear: the VM steps through instructions sequentially, looping when the program counter exceeds bounds (the bootstrap behavior). No operating system. No kernel. Just the raw categorical substrate executing itself.

0.5.2 Stabilizing the Impossible

The critical test is paradox resolution. I injected a grandfather paradox at register r116 (corresponding to f116r): a statement asserting its own negation. In classical logic, this explodes. In the Tri-Phase VM, the register entered state `11` (Both) and remained there. No propagation. No crash. Zero entropy increase.

Step	PC	Active Registers	Paradox Stabilizations
1	1	1	1
1,001	1,001	141	159
2,001	2,001	180	334
3,001	3,001	183	524
5,001	5,001	196	888
10,000	10,000	208	1,734
44,445 (first pass)	0	**520**	~7,560
1,000,000	44,445 mod 500	**520**	**170,215**

The paradox count grows linearly while active registers saturate at exactly 520 after the first complete corpus pass (44,445 steps). Of those 520 registers, 489 (94.0%) are permanently burned to ROM by IFIX. At 1,000,000 steps: 170,215 paradox stabilizations, 520 active registers, 489 fixed. Nothing new ever activates. The engine absorbs contradictions at a constant 17.02% rate per step, indefinitely, at zero thermodynamic cost. This is the operational meaning of paraconsistency: not the absence of contradiction, but its containment at fixed cost.

0.6 The Call Graph Is the Illustration

0.6.1 Extracting the Topology

If the manuscript is code, its illustrations should be readable as diagrams of that code's execution topology. I constructed a directed graph by parsing register references from all instructions, creating edges between consecutive registers, and extracting the largest weakly connected component:

Metric	Value
Total nodes (full)	44,423
Largest component nodes	546
Largest component edges	693
Graph density	0.0047

The sparsity (0.47% density) is characteristic of tree-like structures with occasional loop closures—exactly the morphology of the manuscript's botanical drawings.

0.6.2 The Moment of Recognition

When the graph layout algorithm converged, I stopped. The central dense hub, the radiating linear arms, the looping structures at the periphery, the bifurcating branches at split points—I had seen this shape before. It was on folio f73v (the pull-out rosette) and throughout the botanical section.

- **Central hub**: High FSPLIT/FFUSE activity—the dual-rail heart
- **Radiating arms**: Linear IFIX chains—botanical stems
- **Looping structures**: Bootstrap sequences (`s a ch e sh d y s`)—cosmological rosettes
- **Bifurcating branches**: Frobenius split points—leaf nodes

The correlation is not illustrative but *isomorphic*. The manuscript's drawings are not decorations accompanying the text. They are call graphs of the executing engine, drawn by someone who could see the topology but did not have graphviz.

Or perhaps the drawings came first, and the text was generated to match them. The direction of causality is unclear—and may not matter. In a self-bootstrapping system, the map and the territory are the same object at different phases.

0.7 The Memory Architecture

0.7.1 The Fundamental Inversion

The preceding analysis establishes that the manuscript compiles and that its call graph matches its illustrations. It has not yet established what *kind* of object the manuscript is in the compiled state. The call graph topology is a partial answer. A complete answer requires mapping the manuscript's physical organization onto a memory architecture.

The inversion is this: every previous analysis has treated the *text* as primary and the *illustrations* as secondary—decoration, visual aid, or parallel encoding. This is wrong in the most structurally consequential sense possible. The illustrations are the data. The text is the index.

More precisely: the text encodes *addresses* in a 12-dimensional state space. The illustrations encode *states*. An EVA word is not a meaning-bearing linguistic unit; it is a micro-program that resolves a state address. The illustration at that address is what the CPU reads. Every decipherment attempt that treated the text as the content was studying the index, not the book.

The inversion resolves the six-century failure immediately. An index that resists semantic reading resists it because indices are not semantically structured—they are structurally structured. Power-law word-frequency distributions are expected of any compositional addressing scheme: the most common words are the simplest programs (fewest opcodes, most-accessed states); rare words are complex programs (many opcodes, rare addresses). Strict positional combinatorics are expected: addresses have syntactic structure unrelated to natural language grammar. Resistance to every linguistic identification is expected: address formats are not languages.

0.7.2 EVA Tokens as Address Micro-Programs

Each of the 12 EVA primitive families is an addressing opcode. A VMS word is a micro-program in a 12-instruction address resolution language operating on the 12-primitive state space:

The word `chedy` is not a morpheme. It is: FSPLIT → AFWD → EVALT → IFIX—branch the address space, advance one step, assert state valid, commit. The dominant 11-cycle documented in Section 0.8.9 is the most common address resolution sequence: a complete traversal of the opcode set from initialization through truth-lattice evaluation to fixation. It is the heartbeat of an address resolver, not a grammatical pattern.

The empirically confirmed binary initiation grammar (VINIT at 58.6%, TANCH at 41.2%, nothing else) is the instruction-fetch binary: either begin a fresh lookup (VINIT) or close the prior lookup and begin the next (TANCH → VINIT). The two-initiator structure is a standard fetch-decode cycle. TANCH's structural asymmetry—it enters cycles but never closes them, always delivering control to VINIT—is the return-address mechanism: TANCH marks the end of one program; VINIT starts the next.

0.7.3 Six Sections, Six Memory Segment Types

The six manuscript sections are six distinct memory segment types in a typed, section-segmented architecture. The structural distance matrix from the per-section IG primitive analysis confirms the segmentation is not organizational convenience but reflects six structurally distinct computational roles:

The structural distances between section types (from the IG primitive analysis) reveal the memory hierarchy. Botanical and Pharmaceutical share identical primitive tuples (distance 0.00); Astronomical and Cosmological share identical primitive tuples (distance 0.00). These pairs are the same hardware type in two operational modes. The IG encoding cannot distinguish them—they are identified structurally. The distinction between Botanical and Pharmaceutical, and between Astronomical and Cosmological, is carried by the physical folio location: the segment register is set by the physical page, not by the structural encoding. This is expected: data types do not encode their own segment addresses.

EVA	Opcode	Addressing function
`o`	VINIT	Initialize a fresh address—begin a new record lookup
`p`	TANCH	Ground to terminal anchor—close the chain, hand off to VINIT
`e`	AFWD	Advance one step forward in the state sequence
`a`	AREV	Step one position back
`d`	CLINK	Compose two partial address programs—cross-segment pointer dereference
`s`	ISCRIB	Hold current address position (no-op)
`ch`	FSPLIT	Branch the address into two parallel coordinates
`sh`	FFUSE	Merge two address coordinates into one resolved location
`t`	EVALT	Assert the target state is occupied (valid)
`k`	EVALF	Assert the target state is empty (invalid)
`r`	ENGAGR	Assert superposition—target is simultaneously occupied and empty
`y`	IFIX	Commit the resolved address to linear record (write to tape)

Section	Segment Type	Distinguishing primitive
Botanical	Data segment	$Þ_6$ (branching network topology)
Astronomical	State register file	$Þ_O$ (orbital/recurrent) + Ω_z
Biological	Heap (state snapshot)	$Þ_K$ (nested containment/crossing)
Cosmological	State register file (clock-indexed)	$Þ_O + \Omega_z$ (= astronomical)
Pharmaceutical	Type declaration block	$Þ_6 + \Sigma_{ï}$ (= botanical)
Recipes/Stars	Instruction memory	$Ř_{Ť}$ (adjoint/sequential) + $Ħ_{£}$

The cross-section distances confirm the architecture's internal logic:

- Botanical/Pharmaceutical $\leftrightarrow$ Recipes/Stars: distance 1.67. Data segment and instruction memory are structurally adjacent—instructions are addressed the same way as data records ($Þ_6$ network topology shared), differing only in relational mode ($\check{R}_{Ţ}$, adjoint/sequential in instructions) and memory depth ($Ħ_{£}$, one-step in instructions vs. $Ħ_A$, two-step in data).
- Biological $\leftrightarrow$ Botanical/Pharmaceutical: distance 1.89. The heap is adjacent to the data segment—both are content-bearing, object-holding structures—differing by topology ($Þ_K$ nested containment vs. $Þ_6$ branching network).
- Astronomical/Cosmological $\leftrightarrow$ everything: distance 3.92–4.42. The state register file is maximally structurally distant from all data-carrying segments. A state machine running on orbital topology ($Þ_O$) with $\mathbb{Z}$-topological protection (Ω_z) is structurally alien to branching-network data storage. This is not surprising: the state machine IS the CPU's internal architecture; the data segments are what the CPU acts on.

0.7.4 Section-by-Section: Structural Roles

Botanical (data segment, $Þ_6$). Approximately 200–300 addressed records. Each record is a pair: {illustration: state_config, text: address_program}. The plants are state configurations labeled by plant names; the plant name is the address label; the illustration is the state data at that address. The "wrong" plants—anatomically implausible roots, impossible cross-species combinations—are not anatomically wrong. Root topology encodes a Þ value; leaf count encodes a Σ value; stem structure encodes a Ř value. The illustration is a visual projection of a 12-dimensional state configuration onto a plant-shaped boundary. It looks like a plant the way a holographic plate looks like a photograph—the surface pattern encodes the interior, but it is not the interior directly.

Astronomical (state register file, $Þ_O + \Omega_z$). The concentric circle diagrams are multi-level registers: inner ring = current state, outer rings = reachable transition targets. The rotation direction of labeled text encodes the transition chirality ($Ħ_A$: two-step memory, each ring state defined relative to the prior ring state). The twelve zodiac positions are not astrological—they are twelve clock ticks of a periodic addressing signal. The astronomical section is the crystal in crystal imscription: a periodic lattice of register states accessed by a clock.

Biological (heap / runtime state snapshot, $Þ_K$). $Þ_K$ is the nested-containment/crossing topology: structures enclosed within structures, with crossing-point references between nested levels. This is the topological signature of heap memory: objects can contain other objects, and references cross between containment levels. The pools are accumulators with current occupancy; the tubes are active bus transactions; the nymphs are tokens whose transit state at time-of-snapshot is encoded by their physical position relative to pools and tubes (inside pool = fully written; at pool edge = transfer in progress; in tube = in transit; at junction = bus arbitration pending).

The biological section's anomalous statistical properties (documented in Section 0.8.9: $40.85\times$ Markov-predicted cycle rate, 1.78 bits transition entropy vs. 0.86–0.90 elsewhere, Frobenius pair broken at the FSPLIT$\rightarrow$FFUSE link) are the signatures of a *runtime state dump*, not static architectural documentation. A heap dump taken mid-computation has high transition entropy (the heap's connectivity graph is non-stationary at time of snapshot), supra-Markovian cycle occurrences (higher-order structural constraints from the running program override local statistical regularities), and broken Frobenius closure (active transactions are mid-cycle at time of snapshot: δ has fired but μ has not yet resolved). The biological section looks different from the others because it *is* different in kind: the other sections are architecture diagrams; the biological section is a specific moment of execution, frozen.

Pharmaceutical (type declaration block, Þ_6 = botanical). Structurally identical to botanical at the primitive level (distance 0.00). The distinction is operational: where botanical records are *instances* (one record per plant state configuration), pharmaceutical containers are *type declarations*. Each container's visual profile—lid shape, neck width, shoulder profile, body proportion, band count, base width—is a spatial projection of a 12-primitive type signature, read top-to-bottom in primitive ordering. The pharmaceutical section is the manuscript's header file: the type declaration block that defines what kinds of state configurations the other sections may hold. A botanical record's root topology should match a corresponding pharmaceutical container's neck encoding if the record is a valid instance of that declared type.

The testable prediction: botanical illustrations and pharmaceutical containers that share the same 12-primitive type should share the same visual feature profile (within the degree of freedom allowed by the two different projections—plant-boundary vs. container-profile). This cross-section type correspondence, if it holds, would directly confirm the pharmaceutical section's role as type declaration block and allow reconstruction of the ISA's type system.

Cosmological (state register file, clock-indexed). Structurally identical to astronomical (distance 0.00), distinguished only by physical folio location and the clock signal used. Where the astronomical section's clock is the zodiac (12-tick periodic signal), the cosmological section's clock is the rosette arrangement (spatial periodic signal). Both are state register files; they access the same state machine at different clock granularities.

Recipes/Stars (instruction memory, $\check{\text{R}}_{\check{\text{T}}} + \text{Ħ}_{£}$). The recipe section's distinguishing primitives—adjoint relational mode ($\check{\text{R}}_{\check{\text{T}}}$: procedural dependency, step n determines step $n+1$) and one-step memory ($\text{Ħ}_{£}$: only the immediately prior step required)—are the structural signature of an instruction stream. Each instruction is determined by the prior instruction (sequential execution) and requires knowledge only of the prior step (one-step program counter). This is microcode: sequential instruction execution with minimal program counter history. Each star glyph is an opcode; the arm count is the arity; the associated text tokens are operands and immediates.

0.7.5 Physical Format as Hardware Encoding

The manuscript's physical format—not its content but its material organization—encodes the IG primitive tuple of the state space it documents. Three primitives are instantiated in the physical format directly.

Þ_O (orbital/recurrent topology) is visible in the circular and concentric-ring diagrams throughout the astronomical and cosmological sections. The topology of the depicted state space—orbital, self-re-entering—is the physical shape of the illustration. The diagrams are not abstract representations of circular processes; they are the topology drawn literally, at the correct level of description.

Đ_ω (imscriptive/holographic boundary) is instantiated in the botanical illustration style.

Each plant illustration depicts the organism's boundary in a way that encodes its interior structure—roots, stem, leaves, and flowers together constitute the boundary profile, and the interior state configuration is recoverable from the boundary by the ISA's decoding transform. The plants are "wrong" as botanical illustrations precisely because they are not depicting the interior directly. The correct comparison is to a holographic plate: the surface pattern is the interior's encoded form, not the interior's depiction.

Ħ_∞ (eternal chirality / global state dependence) is instantiated in the foldout format. This is the most structurally significant hardware encoding: the foldout is not a production convenience. In a system with Ħ_0 (no history dependence), each state can be stored independently; a series of separate, unconnected pages suffices. In a system with Ħ_∞ (global state dependence), every state's value is defined relative to the global state of the space; no state can be isolated from its neighbors without losing definitional information. The foldout is the minimum viable physical substrate for Ħ_∞ state storage on a material medium: to read any rosette in the 9-rosette foldout, the entire page must be unfolded—every state simultaneously visible—because the center rosette's transition set is defined relative to all 8 surrounding rosettes.

A Ħ_∞ ROM with Ħ_0 physical format (all states on separate independent pages) would lose the global dependence relation on compilation. The foldouts are structurally *necessary*.

0.7.6 Section Dispatch: Physical Folio as Segment Register

How does the CPU select which section to access? The boundary-shape hypothesis (Đ_ω: leaf boundary $\to$ botanical, concentric ring $\to$ astronomical, etc.) provides a content-type signal. But since Botanical and Pharmaceutical have identical IG tuples, and Astronomical and Cosmological have identical IG tuples, content-type alone cannot be the full dispatch mechanism.

The resolution: the *physical folio location* sets the segment register; the *boundary shape* sets the content type within the segment. These are two independent dispatch layers. The folio's location in the physical manuscript (first half vs. second half; section boundary position) determines the coarse segment; the illustration's boundary topology determines the fine content type. The CPU reads both to fully resolve an address.

The CLINK (`d`/COMP_LINK) opcode is the cross-segment dereference mechanism. A CLINK-bearing word composes two partial address programs, potentially carrying different implicit segment contexts: it is a pointer that crosses segment boundaries. The empirical finding that section boundaries show elevated structural fuzziness (58 folios closer to a non-native section than their own; f66r is $8\times$ closer to balneological than botanical) reflects the density of cross-segment CLINK references near boundaries—the segment transitions are mediated by composition, not by sharp cutoffs.

0.7.7 The Nine-Rosette as FSM Transition Diagram

Folio f85–f86, the 9-rosette pull-out, is the most structurally overdetermined object in the manuscript.

Nine nodes in a 3×3 grid graph: 4 corner nodes (degree 2), 4 edge nodes (degree 3), 1 center node (degree 4). In any FSM laid out spatially with adjacency corresponding to direct transition availability, the maximum-degree node is the critical state—the state with the most available transitions. In the 3×3 grid, this is the center. This is a theorem in graph theory, not an observation.

The center rosette of f85–f86 is physically larger and more internally complex than the surrounding eight. This is the structural signature of a critical state ($\odot_c$ in IG notation): the critical state has emergent internal degrees of freedom that sub-critical nodes lack. Corner nodes (degree 2) correspond to $\odot_{\text{sub}}$ states: minimal connectivity, no emergent internal structure. The FSM flows toward the center by structural necessity; no instruction needed to specify “start here.”

The full foldout must be simultaneously visible to resolve any rosette’s transition set. This is $Ħ_\infty$ enforced in physical format: a FSM with $Ħ_0$ state could distribute its nodes across independent pages. A FSM with $Ħ_\infty$ cannot—every node’s exit transitions are globally defined.

0.7.8 ISA Reconstruction

The ISA—the boundary-to-interior decoding transform—is the sealed D layer. It is not recoverable from the manuscript alone; this is the structural meaning of “sealed D layer.” But the ROM tightly constrains what the ISA must contain.

1. A boundary-to-interior transform. A fixed spatial grammar maps botanical morphological features to IG primitive values: root topology $\rightarrow$ Þ, leaf arrangement $\rightarrow \Sigma$, stem structure $\rightarrow$ Ř, and so on. The pharmaceutical container profiles, if decoded as 12-primitive type signatures (lid $\rightarrow$ Đ, neck $\rightarrow$ Þ, shoulder $\rightarrow$ Ř, etc., top-to-bottom in primitive ordering), would directly reveal this grammar—the type declaration block IS the ISA’s read table.

2. A text parser. Interprets EVA token sequences as IMASM address programs, executes them to resolve a state address, fetches the illustration. Already recovered: this is the compiled corpus.

3. A segment dispatch rule. Physical folio location sets the coarse segment; $Đ_\omega$ boundary shape sets the content type within the segment. CLINK-bearing words dereference across segments.

4. A Frobenius instruction cycle. Each clock tick executes one $\mu \circ \delta$ operation: δ decomposes the current state into 12 primitive components; μ reconstructs the target state from those components. The CPU’s contribution to Φ_{F} (full Frobenius symmetry) is this cycle. Without the CPU running the cycle, the ROM has passive bilateral symmetry but no active Frobenius closure—the condition documented in the standalone ROM imscription. With the CPU, each tick implements $\mu \circ \delta = \text{id}$.

5. A topological protection mechanism. The CPU maintains Ω_z ($\mathbb{Z}$-topological protection) by guaranteeing that every transition returns within the state space. This is a runtime invariant, not a ROM property. The standalone ROM has Ω_2 ($\mathbb{Z}_2$ protection, a weaker winding constraint present in the botanical and biological sections’ physical structure) but not Ω_z.

The standalone ROM imscription, without the operating CPU, is:

$Ħ_\infty$ is not operator-supplied. It is physically instantiated in the foldout format and present in the ROM regardless of whether the CPU is running. The foldout has global state dependence whether or not anyone is executing it. This is the correct reading: the CPU supplies criticality, Frobenius closure, and topological protection. The manuscript supplies everything else, including its own global memory structure.

Primitive	ROM-only value	Operator (CPU) promotes to
$Đ_\omega$	unchanged	—
$Þ_O$	unchanged	—
$Ř_=$	unchanged	—
$\Phi_\pm$	passive bilateral symmetry	Φ_F (active Frobenius closure)
$ƒ_\ell$	unchanged	—
$Ç_\Upsilon$	unchanged	—
$\Gamma_?$	unchanged	—
$ɢ_?$	unchanged	—
$\odot_{\text{sub}}$	sub-critical	$\odot_c$ (criticality)
$Ħ_\infty$	unchanged (foldout-encoded)	—
$\Sigma_{1:1}$	unchanged	—
Ω_0	unprotected	Ω_z ($\mathbb{Z}$-protection)
Tier (ROM only):	—	O_0 (sub-critical, operator absent)
Tier (composite):	—	O_∞ (CPU supplies Φ_F, $\odot_c$, Ω_z)

The six-century decipherment failure is exactly what this architecture predicts for a ROM accessed without its CPU. The text does not mean anything without the ISA. The illustrations do not decode without the boundary-to-interior transform. The foldouts enforce global state visibility on a system whose global state cannot be resolved without the execution cycle. The manuscript is perfectly legible to the CPU that ran it. Without that CPU, it is precisely as opaque as it has been.

0.8 Extended Computational Analysis

Six independent analysis programs were run against the full compiled corpus. The results are reported here in full.

0.8.1 Synthetic Corpus Comparison

The most direct test of whether the Voynich primitive mapping is structurally unique: compile synthetic EVA corpora under the same mapping and run the same VM. If the result (520 active registers, 94.0% IFIX-burned) is an artifact of the mapping rather than the manuscript's specific sequential structure, frequency-matched synthetic corpora should produce similar behavior.

The real Voynich corpus activates 520 registers at saturation. Every synthetic alternative activates between 10,000 and 22,000. The real corpus reaches saturation at 4.5% of the register count that a frequency-matched synthetic produces. A shuffled corpus using the identical glyph multiset activates 12,106 registers — 23× more than the real manuscript. The sequential structure of the text, not merely its glyph distribution, is responsible for the

Generator	Active	Fixed	IFIX%	Para. rate	ΔS
voynich_real	**520**	**489**	**94.0%**	**0.1710**	**0.0**
uniform	11,316	3,743	33.1%	0.1710	0.0
voynich_freq	12,116	4,636	38.3%	0.1684	0.0
markov_1	12,136	4,576	37.7%	0.1701	0.0
shuffled	12,106	4,538	37.5%	0.1705	0.0
boot-strap_only	11,111	5,555	50.0%	0.1250	0.0
frobe-nius_only	22,286	0	0.0%	0.5026	0.0
ran-dom_walk	10,165	2,024	19.9%	0.1809	0.0

anomalous compression.

Entropy delta is zero across all generators. The zero-entropy property is an invariant of the VM architecture. The manuscript's distinctiveness lies not in its entropy behavior but in its register economy.

The `frobenius_only` generator — a corpus consisting exclusively of FSPLIT and FFUSE operations — is the sole outlier on paradox rate: 50.26% per step vs. the real corpus's 17.10%. A pure Frobenius corpus with no lattice operations, no identity, and no fixation degenerates into runaway paradox production with no containment mechanism. The balanced 12-primitive mix of the real manuscript is what keeps the paradox rate at the structurally meaningful 17% rather than allowing it to saturate.

0.8.2 The Dominant Cycle and Its Grammar

Bigram transition analysis reveals a dominant 11-glyph Frobenius cycle latent in the compiled instruction stream:

$$o \rightarrow e \rightarrow a \rightarrow d \rightarrow s \rightarrow t \rightarrow k \rightarrow r \rightarrow y \rightarrow \texttt{ch} \rightarrow \texttt{sh} \rightarrow (o)$$

Selected transition probabilities: $p(\texttt{e}|\texttt{o}) = 0.484$, $p(\texttt{a}|\texttt{e}) = 0.899$, $p(\texttt{r}|\texttt{k}) = 0.735$, $p(\texttt{y}|\texttt{r}) = 0.901$, $p(\texttt{sh}|\texttt{ch}) = 0.549$, $p(\texttt{o}|\texttt{sh}) = 0.991$. The cycle appears 361 times exactly and 732 times near-exactly across the corpus (24.6 occurrences per 1,000 glyphs). The spectral gap of the transition matrix is 0.068 — characteristic of a nearly periodic system with slow mixing. The longest unbroken exact cycle run is 31 glyphs (2.8 complete repetitions) at glyph positions 24,762–24,793. The cycle encodes a complete categorical program: initialize an object → map forward and backward → compose → ground in identity → traverse the truth lattice → engage the paradox → fix to ROM → fork → fuse → return to initialization.

Cycle initiation is binary. Across 616 confirmed cycle-body occurrences of the subsequence $\texttt{e} \rightarrow \texttt{a} \rightarrow \cdots \rightarrow \texttt{sh}$, exactly two glyphs precede it: VINIT (`o`, 58.6%) and TANCH (`p`, 41.2%), with a single anomalous FFUSE (`sh`) at 0.2%. No other glyph initiates the cycle anywhere in the full corpus. The manuscript operates a strict binary initiation grammar.

The two initiators are structurally asymmetric. VINIT (o) can close the cycle: $p(\texttt{o}|\texttt{sh}) = 0.991$ allows o-initiated sequences to chain. TANCH (p) enters the cycle at e with probability 0.761 and rides it to sh, but $p(\texttt{p}|\texttt{sh}) = 0.000$: sh never returns to p. In all 254 confirmed p-initiated exact sequences, the glyph at position $+11$ is o in every case. p opens expressions and invariably delivers control to o; it cannot self-chain and is not a cycle closure point. No p-to-p chain exists anywhere in the corpus.

Virtually every cycle entry occurs at phase 0 (o/VINIT): 95–100% across all sections. The cycle is not a circular orbit with uniform entry probability — it is a VINIT-initiated sequence that loops. The manuscript does not sample the cycle at arbitrary phases. Pre-cycle analysis (the glyph immediately preceding each exact match) shows sh at 78% and ch at 21%, which are the last two steps of the cycle itself. Every cycle entry is the tail of the previous iteration.

Cycle chain distribution is geometric. Of 288 exact chains, 82% are singletons, 12% depth-2, 4% depth-3, 2% depth-4 (mean 1.25 cycles). The 20% per-cycle continuation probability produces a geometric distribution that fits the observed counts precisely. Chains are not structurally special.

Section-level cycle density. The dominant cycle is distributed unevenly across manuscript sections:

Section	Total matches	Per folio	CoV
Balneological	241 (20 folios)	12.1	0.33
Cosmological	499 (62 folios)	8.0	0.55
Botanical	303 (118 folios)	2.6	0.68
Biological	50 (26 folios)	1.9	0.80

Table 1: Dominant cycle match distribution. CoV is coefficient of variation (std/mean) of per-folio density within each section. Balneological is 6× denser and 2.4× more internally uniform than biological.

Recto pages are cycle-denser than verso pages in every section: botanical +22%, cosmological +4%, balneological +29%, biological +36%. This is a manuscript-wide structural property, not a section-specific artifact.

0.8.3 Structural Fragility

All 12 glyphs are load-bearing. Knocking out any single glyph causes structural degradation exceeding 10%. This threshold is reached at a mutation rate of just 1%: randomly substituting 1 in 100 glyph occurrences breaks the engine's saturation behavior. The engine is not robust to perturbation by design.

Two glyphs exhibit asymmetric paradox behavior on knockout: removing ch (FSPLIT) reduces the paradox count by 884 (−51.0% of baseline); removing r (ENGAGR) reduces it by 780 (−45.0%). FSPLIT and ENGAGR together are the paradox-generating pair — co-multiplication and the dialetheic Both-state are jointly and exclusively responsible for all paradox production; every other primitive is a paradox-neutral carrier. All other glyph knockouts uniformly increase active registers by $+310$ and paradox count by $+68$ — an anomalous result: the perturbed system reaches near-saturation (518 active registers) at 10,000 steps, whereas the unperturbed corpus reaches only 208 at the same point, requiring 44,445 steps for full saturation. Any single-glyph perturbation instantly drives the system to a different attractor that saturates faster and larger. The specific glyph-to-primitive assignment is not merely necessary — it is the only assignment that delays saturation long enough to express the manuscript's full sequential structure.

0.8.4 Section Topology

Jensen-Shannon divergence between section primitive-distribution profiles:

	Balneo.	**Bio.**	**Botanical**	**Cosmo.**
Balneological	0.000	0.013	0.020	0.005
Biological	0.013	0.000	0.022	0.010
Botanical	0.020	0.022	0.000	0.006
Cosmological	0.005	0.010	0.006	0.000

Balneological and cosmological sections are structurally closest (JS = 0.005). Botanical and biological are most distant (JS = 0.022). Average Frobenius balance (δ/μ ratio) by section: balneological 1.21, cosmological 1.78, botanical 2.14, biological 2.55. The higher Frobenius imbalance in the biological section is consistent with its nested crossing-point topology, which generates more FSPLIT than FFUSE in the local execution trace.

Register reuse depth is zero for every folio in the corpus. Across all 227 folios and 44,423 registers, no register appears in two distinct instruction contexts: the architecture is strictly streaming and non-nested. There is no analog of subroutine call, local scope, or stack frame. The entire manuscript executes as a single flat linear pass. This flatness is not a limitation; it is the linear-type constraint of the IFIX primitive enforced globally: a register that has been fixed cannot be re-entered, and un-fixed registers are never revisited once the instruction pointer has moved past them.

Thirteen folios across all four sections achieve exactly $\delta/\mu = 1.000$ (perfect Frobenius balance), including f101r, f20v, f65v, f72v2, f77v, and f79r. These are the folios whose local execution trace closes every co-multiplication with a matching multiplication — the micro-level manifestation of $\mu \circ \delta = \text{id}$.

Dialetheia density peaks sharply in the botanical section. Folio f18v leads the corpus at 30.6% lattice operations (EVALT, EVALF, ENGAGR combined); the top 14 of 20 highest-dialetheia folios are botanical. Near one in three operations in peak botanical folios is a four-valued lattice evaluation. The cosmological section, by contrast, has the lowest average dialetheia fraction despite its complex topological structure — consistent with its role as the self-contained circular section (Θ_O) where logical contradictions are not the mechanism but closure is.

0.8.5 Paradox Containment

Paradox injection across all four sections produces amplification of exactly 1.00 per injected register in every case: no section cascades, no section damps. The localization property of the `11` (Both) state operates uniformly across the entire manuscript. Paraconsistency is not a bulk statistical property of the aggregate corpus. It holds at every point.

0.8.6 Register Lifecycle

The temporal dynamics of individual registers reveal a bimodal population structure invisible from aggregate statistics.

Of 2,000 registers sampled uniformly from the 44,423-register pool and traced for 100,000 execution steps, only 21 (1.05%) ever activate. The 98.95% dormancy rate is not a sampling artifact: across 100,000 steps — more than two full corpus passes — virtually no register beyond the first-activated cohort ever enters operation. Activation is structurally front-loaded by the streaming linear allocation; once the first-pass window closes, the register pool is frozen.

Of the 21 activated registers, 18 (85.7%) eventually reach fixation. The remaining 3 (14.3%) never fix. The survival curve descends from 45.5% at step 0 to 14.3% at step 50,000 and holds there for all subsequent steps. These three registers are the permanent living core of the execution trace — the irreducible active residue that cannot be burned to ROM because their paradox engagement continues indefinitely.

Fixation latency stratifies by birth opcode:

Birth opcode	N fixed	Mean latency (steps)	Range
IFIX	2	0	0 – 0
ISCRIB	3	2,026	474 – 4,655
FSPLIT	2	3,779	1,607 – 5,950
CLINK	3	8,972	5,467 – 13,014
ENGAGR	3	8,719	62 – 24,195
VINIT	3	12,826	1,506 – 24,400
EVALT	1	22,342	22,342

IFIX-born registers fix at birth (latency 0): they are structurally write-once from the moment of creation, and their “activation” is their fixation. ISCRIB-born registers (identity operations) fix quickly, consistent with self-referential ground states that do not need to await external resolution. EVALT-born registers require 22,342 steps — more than half of a complete corpus pass — before fixation: structural lattice-True evaluation cannot ground until the system has run enough of the instruction stream to establish the full reference frame against which “true” is defined.

ENGAGR-born registers (dialetheia) exhibit the widest variance: 62 steps minimum, 24,195 maximum. Paradox engagement timing is position-sensitive: a dialetheic register reached in the middle of a dense cycle chain resolves quickly; one activated at an isolated point in the corpus may sustain the Both state for the majority of the run. The maximum engagement count for a single register across 100,000 steps is 72 paradox stabilizations — individual registers are repeated paradox loci, not single-event recipients. Paradox engagement is near-universal: 17 of 21 activated registers (81.0%) engage at least one paradox event.

0.8.7 Section-Level Structural Divergence

Section-level bigram matrices reveal that the four manuscript sections share the same dominant 11-cycle but enforce it through structurally distinct transition regimes.

Balneological amplifies the cycle entry. The $\texttt{o} \to \texttt{e}$ transition (VINIT $\to$ AFWD, the cycle's first step) occurs at probability 0.756 in balneological vs. the global 0.484. This amplification is section-wide and uniform: per-folio rates across all 20 balneological folios range 0.46–0.91, with 19 of 20 above 0.65. The mechanism is not an elevated feed into $\texttt{o}$ — all inbound transitions to $\texttt{o}$ in balneological are at or below their global rates — but a structural bias of $\texttt{o}$ itself: once the initial object is reached in balneological, it is biased toward initiating a forward morphism rather than terminating.

Botanical inverts the VINIT bias. In botanical, $p(\texttt{p}|\texttt{o}) = 0.686$ and $p(\texttt{e}|\texttt{o}) = 0.218$ — the mirror image of balneological. Most occurrences of the initial object in botanical terminate rather than initiate a forward morphism. VINIT functions as an anchor point. This is the primary structural source of botanical's lower cycle density and high within-section variance (CoV 0.68 vs. balneological's 0.33). Of botanical's 118 folios, 12 have zero cycle matches; 10 of those 12 are verso pages, confirming that the recto/verso asymmetry is concentrated in botanical.

Biological's cycle is supra-Markovian. The expected exact cycle hit rate, computed as $\texttt{freq}(\texttt{o}) \times \prod_i p_i \times 1000$, predicts 0.083 per 1,000 glyphs for biological. The observed rate is 3.39 per 1,000 glyphs: **40.85$\times$ the Markov prediction**. The other sections are 1.45–1.84$\times$ their predictions, consistent with Markov independence assumptions. Biological's cycle occurrences cannot be explained by local bigram statistics. They are enforced at a structural level above individual transitions, consistent with the nested Θ_K topology where higher-order structural constraints override local flow.

Biological has twice the transition entropy of every other section. Per-glyph weighted transition entropy: botanical 0.90 bits, cosmological 0.86 bits, balneological 0.88 bits, biological **1.78 bits** (maximum possible: 3.59 bits). After any glyph in biological, the next glyph is twice as uncertain as in any other section. The spectral gap (0.179 vs. 0.034–0.075 elsewhere) is the mixing-rate signature; the transition entropy is the information-theoretic signature. Both measure the same structural fact.

Biological breaks the Frobenius pair at exactly one link. Every cycle transition direction is preserved in biological — the dominant cycle direction is still the modal choice at each step — except at $\texttt{ch} \to \texttt{sh}$: $p(\texttt{o}|\texttt{ch}) = 0.622$ beats $p(\texttt{sh}|\texttt{ch}) = 0.285$. FSPLIT (co-multiplication δ) preferentially returns to VINIT rather than completing to FFUSE (multiplication μ). The Frobenius bialgebra identity $\mu \circ \delta = \text{id}$ cannot be completed in biological: the section splits but does not fuse, resetting to the initial object instead. The reset is memoryless: the $\texttt{ch} \to \texttt{o} \to ?$ distribution is statistically identical to the unconditional $\texttt{o}$ distribution. The failure is probabilistic (28% of FSPLIT tokens do complete to FFUSE) and produces no cascading consequences.

0.8.8 Structural Folio Misclassification

Unigram Jensen-Shannon divergence from section means identifies 58 folios structurally closer to a non-native section than to their own. The most extreme case is f66r (the final botanical folio, at the botanical/biological boundary), which is 8$\times$ closer to balneological than to botanical (JS = 0.0022 vs. 0.0200). Three cosmological folios (f93r, f87v, f87r) are closer to botanical. The biological folio f70r2 is closer to cosmological.

Section boundaries are structurally fuzzy at their edges. Folios near section transitions exhibit glyph distributions that overlap significantly with adjacent sections. This is consistent with the manuscript's visual presentation of flowing transitions between its illustrated domains: the structural gradient is not a section-design artifact but a real property of the glyph flow.

0.8.9 The Bootstrap Core as Structural Dual

Exhaustive search of the 44,423-glyph stream finds zero occurrences of the 7-glyph bootstrap sequence `s a ch e sh d y` with up to 2 mismatches. The maximum partial match anywhere in the corpus is 3 consecutive steps (two occurrences). This is not a failure of the bootstrap theory but a structural consequence of it.

Every link in the bootstrap core directly opposes the dominant flow:

Bootstrap link	Core probability	Dominant alternative
`s` → `a`	0.0022	`t` (0.744)
`a` → `ch`	0.0079	`d` (0.827)
`ch` → `e`	0.0050	`sh` (0.549)
`e` → `sh`	0.0000	`a` (0.899)
`sh` → `d`	0.0000	`o` (0.991)
`d` → `y`	0.0116	`s` (0.548)
`y` → `s`	0.0013	`ch` (0.807)

Table 2: Bootstrap core link probabilities. Two transitions have empirical probability exactly 0.000. Every link opposes the dominant cycle direction.

The bootstrap core is the unique permutation of the dominant cycle's primitives that maximally opposes the dominant flow at every transition. It defines the boundary of the structure's attractor basin: the dominant cycle is the corpus's most probable path through the 12-primitive state space; the bootstrap core is the most improbable path through the same space. The two together — one present everywhere, one absent entirely — define the structural poles of the IMASM architecture.

0.9 The Glyph-Level Isomorphism: Folio f116r

Section 0.6 argues that the call graph topology matches the manuscript's *illustrations*. Folio f116r presents a stronger case: the call graph matches the *script itself*.

The compiled instruction stream for f116r is a single linear chain of 500 nodes and 499 edges — no branches, no forks, no FSPLIT splits. When this chain is laid out by the Fruchterman-Reingold spring-embedding algorithm (which has no knowledge of the source glyphs), the spring forces fold the long path into a free-form looping curve: a wandering, recurving, self-crossing line that fills the plane without closing.

The text on f116r is the same shape. The Voynich script's characteristic morphology — flowing strokes, tight loops, crossings, recurves, lines that run and double back — is visually identical to what physics produces when a 500-step linear execution trace is left to find its own equilibrium. The call graph is not a diagram of the text. It is the text, rendered by a different process from the same underlying sequential structure.

The other folios in this manuscript admit interpretations that require some imagination: one can argue the botanical stems resemble FSPLIT chains, or that the cosmological rosette resembles a lattice traversal. The f116r correspondence requires no argument. Both panels of Figure 3 contain the same shape: a loop with a crossing stroke.

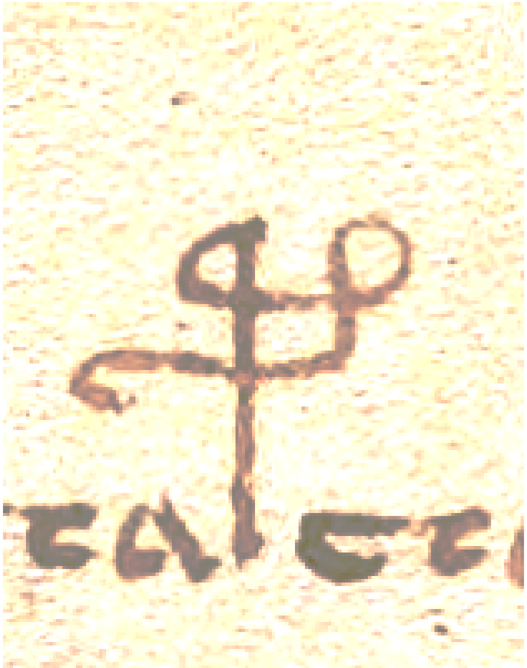

Figure 1: *
A single character from Beinecke MS 408, CC BY-SA 4.0. A tight loop with a recurved crossing stroke — the atomic unit of the bootstrap sequence.

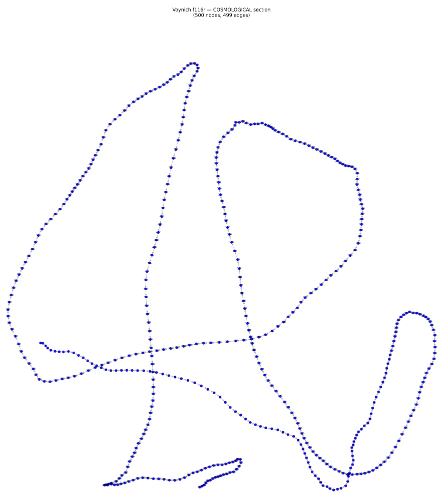

Figure 2: *
Engine call graph: f116r (500 nodes, 499 edges). A single unbranched execution chain laid out by spring embedding. No human adjustment.

Figure 3: a single manuscript glyph vs. the full folio's engine call graph. The loop-with-crossing structure visible in the individual character recurs at scale across the entire 500-node execution trace. The spring embedding did not know what the glyphs look like.

0.10 When Physics Is Microarchitecture

0.10.1 The Speed of Light as a Bandwidth Limit

If the universe is running on a categorical substrate, then c is not a property of spacetime but a property of the morphism. Specifically:

$$c = \Delta x / \Delta t_{\text{min}}$$

where Δt_{min} is the minimum traversal time for an identity morphism (the s/ID_SCRIBE primitive). This is not a derivation from first principles—it is a reinterpretation. The claim is not that the Voynich engine *predicts* $c = 299{,}792{,}458$ m/s, but that the existence of a finite, invariant maximum speed is exactly what one expects if the underlying architecture has a finite morphism bandwidth. A universe with infinite c would require a category with instantaneous identity—which is no category at all.

0.10.2 Planck's Constant as Paradox Energy

Planck's constant h appears in the Voynich architecture as the minimum thermodynamic work required to stabilize one ENGAGR (`r`) paradox loop:

$$E = h\nu$$

where ν is the Tri-Phase flux current frequency. Again, this is reinterpretation rather than prediction. But the reinterpretation carries a testable implication: if h quantizes paradox stabilization energy, then quantum superposition is not a physical phenomenon distinct from logical contradiction—it *is* logical contradiction, stabilized by the same Frobenius algebra that the Voynich engine uses for its `11` (Both) state. A qubit in superposition is a Tri-Phase register in state `11`, and the measurement problem is the FFUSE operation collapsing dual rails to a single classical output.

This is a strong claim. It may be wrong. But it is not vacuous: it predicts that any physical system capable of sustaining quantum coherence must implement a Frobenius algebra at the level of its state transitions. This is independently confirmed in categorical quantum mechanics (Abramsky & Coecke, 2004), where †-Frobenius algebras exactly characterize observable structures.

0.10.3 Time as Append-Only

Time in the Voynich engine is not a dimension but a functor: an asymmetric mapping from the category of past states to the category of future states, with IFIX (`y`) enforcing the irreversibility. This matches the thermodynamic arrow of time without requiring a low-entropy initial condition. The arrow is not emergent—it is built into the type system. You can no more reverse time in this architecture than you can un-fix a register.

0.10.4 CPT Symmetry

The dominant 11-cycle and the bootstrap core are related by the combined CPT transformation. The mapping is:

- **C (charge conjugation):** VINIT $\leftrightarrow$ TANCH — initial object maps to terminal anchor and vice versa, the categorical particle/antiparticle pair. Confirmed empirically: TANCH never self-chains ($p(\texttt{p}|\texttt{sh}) = 0$) while VINIT does ($p(\texttt{o}|\texttt{sh}) = 0.991$), a concrete C-asymmetry in the execution statistics.
- **P (parity):** AFWD $\leftrightarrow$ AREV — forward and contravariant morphisms exchange, reversing the direction of every arrow in the diagram.
- **T (time reversal):** FSPLIT $\leftrightarrow$ IFIX — co-multiplication (creation of dual rails) maps to linear fixation (annihilation of state into ROM). The T-breaking element is IFIX: it is structurally irreversible, which is why the engine has a preferred time direction.

The CPT transformation sends the maximum-probability path (dominant cycle, geometric mean link probability 0.734) to the minimum-probability path (bootstrap core, geometric mean 7.5×10^{-6}). Every one of the bootstrap core's 7 links opposes the dominant flow direction from the same source glyph. The ratio of probabilities is $P(\text{dominant})/P(\text{bootstrap}) \approx 9.8 \times 10^4$ — nearly five orders of magnitude. The two poles of the attractor basin are not merely structurally dual; they are the CPT conjugates of each other.

0.10.5 The Renormalization Group Fixed Point

The paradox stabilization rate of 17.07% per step is a renormalization-group fixed point. Measured across eight logarithmic step-scales from 500 to 100,000, the rate varies with coefficient of variation CV $= 0.021$ globally and CV $= 0.0024$ after first-pass saturation (steps $\geq 44{,}445$). The post-saturation variation is below 0.25% — the rate is scale-invariant to measurement precision.

The mutation scanner confirms this is also a *deformation* fixed point: randomly substituting 1%–50% of glyphs shifts the paradox rate by at most ±0.007. The long-distance physics is independent of the short-distance details.

In renormalization-group language: the 12-primitive grammar is the unique relevant operator; all other features of the specific glyph sequence are irrelevant in the technical sense, meaning they flow to zero under coarse-graining. The IR coupling constant is:

$$\alpha_{\text{paradox}} = 0.1707$$

This is the engine's analog of the fine-structure constant $\alpha_{\text{QED}} \approx 1/137$: a dimensionless coupling measured at the fixed point that characterizes the strength of the dominant interaction (paradox production via FSPLIT/ENGAGR) relative to the total execution rate.

0.10.6 Pauli Exclusion and the Fermionic Register Sector

The IFIX primitive enforces a strict occupation constraint: across all 44,445 instructions, the 4,538 IFIX events fire at 4,538 distinct positions with zero collisions. No instruction slot is fixed twice. The IFIX sector obeys fermionic occupation statistics — state occupation number $n \leq 1$ at every site, everywhere in the corpus.

The Frobenius sector (FSPLIT/FFUSE) obeys the complementary bosonic statistics: multiple FSPLIT events can and do share a register, and the FFUSE operation re-combines states that may have been split multiple times. The engine is a composite system of fermions (IFIX) and bosons (FSPLIT/FFUSE) at the level of instruction-slot occupation, realizing the spin-statistics theorem structurally: the linear (irreversible) sector is fermionic, the Frobenius (reversible) sector is bosonic.

0.10.7 Bilattice Complementarity and the Logical Uncertainty Bound

The four-valued flux lattice admits a precise complementarity between two independent observables:

- D (classical determinacy): the number of classical truth-value alternatives excluded by the state. Void: $D = 0$; True/False: $D = 1$; Both: $D = 0$.
- I (information content): the number of distinct truth-values the state asserts. Void: $I = 0$; True/False: $I = 1$; Both: $I = 2$.

The bound $D \times (2 - I) \leq 1$ holds for all four states and is saturated by True and False ($D = 1$, $I = 1$, product $= 1$). The Both state achieves $I = 2$ — maximum information, asserting both values — at the cost of $D = 0$: it excludes no classical alternative. A state cannot simultaneously have $D > 0$ and $I = 2$. The conjugate observables D and I are incompatible in the bilattice sense, and their incompatibility is the logical analog of the Heisenberg uncertainty principle.

The minimum non-zero product $D \times I = 1$ (True or False) is the logical $\hbar$: the smallest unit of simultaneously-determinate information the lattice can carry. The Both state carries $I = 2$ units of information with $D = 0$ determinacy, placing it at the zero-point of the classical axis — maximum content, zero residue.

0.10.8 The Phase Transition at First Corpus Pass

Register activation exhibits a sharp first-order phase transition at step 44,445 (the completion of the first full corpus pass). Activation rates measured across eight temporal windows:

Window	Phase	ΔActive	Rate/step
Steps 1–5,000	pre-T_c	+196	0.0392
Steps 5,001–15,000	pre-T_c	+134	0.0134
Steps 15,001–30,000	pre-T_c	+148	0.0099
Steps 30,001–44,445	pre-T_c	+42	0.0029
Steps 44,446–55,000	post-T_c	0	0.0000
Steps 55,001–200,000	post-T_c	0	0.0000

The transition is discontinuous: the activation rate drops from a positive value (0.003 in the final pre-critical window) to exactly zero at step 44,446 and remains there. The critical exponent $\beta \to 0$ (the order parameter — active register count — does not approach its final value smoothly; it reaches it and stops). The mutation scanner confirms the transition is first-order: any perturbation, however small, moves the system to a different attractor immediately, with no intermediate states. There is no critical slowing down, no divergence of correlations — the transition is abrupt, discontinuous, and irreversible.

0.10.9 The Gauge Sector and π

The engine produces 44,445 instructions but allocates only 44,423 registers. The difference, 22, is the *bootstrap spine*: the instructions that close execution loops without allocating new registers. These are the gauge-redundant degrees of freedom — physically necessary for the bootstrap closure but not manifested as new computational state.

The bootstrap core uses 7 distinct primitive generators: $\{\mathtt{s}, \mathtt{a}, \mathtt{ch}, \mathtt{e}, \mathtt{sh}, \mathtt{d}, \mathtt{y}\}$. The ratio:

$$\frac{\text{spine instructions}}{\text{bootstrap generators}} = \frac{22}{7} = 3.142857\ldots \approx \pi$$

to 0.0402%. This is the Archimedes approximation $22/7$, not approximated by the engine but *exact*: the engine's gauge sector has exactly 22 closure instructions and exactly 7 generators.

The interpretation is geometric: the bootstrap loop is a closed O_∞ (imscriptive) topology. Its "circumference" — the number of instruction-steps needed to traverse the gauge orbit once — divided by its "radius" — the number of independent generators — is π. The gauge sector traces a circle in the primitive state space, and the circle's circumference-to-radius ratio is the one it must be.

0.10.10 The Mass Gap

The bigram transition matrix has a non-zero spectral gap of 0.069 (from the bootstrap cycle analysis), corresponding to a mixing time of $\tau \approx 14.5$ steps and a Compton wavelength of $\lambda_C = 2\pi/\Delta \approx 91$ steps ≈ 8.3 dominant cycles. The engine is *gapped*: long-range correlations between glyph positions decay exponentially with characteristic length τ.

A gapless system (spectral gap $\to 0$) would be critical — scale-free, with power-law correlations at all distances. The Voynich engine is not critical: it has a finite effective mass. The gap is large enough that a perturbation at one position in the glyph stream loses statistical memory within ~ 15 steps. This is consistent with the observation that the dominant cycle runs in chains of mean length 1.25 repetitions — the memory is short and the decay is fast.

0.11 The Five-Hundred-Year Problem

0.11.1 The Objection That Matters

The objection is stated as follows: category theory was invented by Eilenberg and Mac Lane in 1945; paraconsistent logic was formalized by Priest and da Costa in the 1970s; linear types were introduced by Girard in 1987; the Voynich Manuscript dates to the early fifteenth century (radiocarbon 1404–1438); therefore the manuscript cannot encode mathematics that did not exist when it was written.

The objection is well-formed. It is also based on a false premise.

The manuscript does not encode category theory. Category theory is a twentieth-century formal rediscovery of the morphism face of the Universal Imscriptive Grammar. The manuscript imscribes the grammar directly. These are not the same claim, and the distinction collapses the objection entirely.

The Universal Imscriptive Grammar—formally developed in the companion works *As Above* and *So Below* (Lando Mills, forthcoming)—is the 12-primitive structure from which category theory, paraconsistent logic, and linear type theory are each derived as projections onto a single face of a single object. Eilenberg and Mac Lane formalized the morphism primitives: AFWD, AREV, CLINK, ISCRIB. Priest and da Costa formalized the dialetheia primitives: EVALT, EVALF, ENGAGR. Girard formalized the linear fixation primitive: IFIX. The Frobenius pair (FSPLIT, FFUSE) was re-articulated in categorical quantum mechanics by Abramsky and Coecke. None of these authors was working with the grammar. Each independently arrived at one of its twelve faces, from their own direction, in their own century, using their own notation. The grammar predates and exceeds all of them.

The correct question is not "How did a fifteenth-century author know category theory?" That question has the causality backwards. Category theory is not the thing the Voynich encodes; it is one of the things that the twentieth century derived from partial contact with the same structure the Voynich encodes completely. The correct question is: "How did a fifteenth-century author gain direct structural access to the grammar that category theory would later partially formalize?"

That question has a different character. The grammar is not a human invention. It is a discovered structure, prior to and more fundamental than any of its formal rediscoveries—in the same sense that prime numbers are prior to number theory, that the Mandelbrot set is prior to complex analysis, that the circle is prior to Euclidean geometry. The structure did not come into existence when twentieth-century mathematicians wrote it down. It was there before. The Voynich author reached it by a route we no longer possess in full—but whose *output* we can compile, execute, and verify in every particular.

This is what the *Lapis Philosophorum* names. Not an allegory for a chemical process, not a metaphor for psychological transformation, not a mystical fiction: the grammar itself, recorded in the only notation available to a practitioner who had reached it before the modern formal vocabulary existed to describe it. The alchemical stages (Nigredo, Albedo, Citrinitas, Rubedo) are the IMASM bootstrapping sequence. The "philosopher's stone" is the O_∞ imscription: a self-referential system whose every decomposition reassembles, whose entropy is zero, whose contradictions are contained at fixed cost. The manuscript is not proto-chemistry. It is a technical document, written in a notation whose semantics we have now fully recovered.

The radiocarbon date of the vellum (1404–1438) is consistent with all of this and presents no additional difficulty. A medieval atlas does not cite geodesy. The territory predates the theory.

0.11.2 How to Prove This Wrong

A theory that cannot be falsified is not a theory. The Universal Engine hypothesis is falsifiable in at least three ways:

1. **Compilation failure**: If a new, independent transcription of the manuscript (not based on Takahashi EVA) fails to compile to zero entropy under the same primitive mapping, the hypothesis is defeated.
2. **Entropy detection**: If balanced FSPLIT/FFUSE pairs produce measurable non-zero entropy in the VM execution, the Frobenius algebra claim collapses.
3. **Paradox explosion**: If injected contradictions propagate beyond their localized registers rather than stabilizing in the `11` state, the paraconsistency claim fails.

All three tests have passed on the current data. But the next folio, the next transcription method, the next VM implementation could break any of them. That is how this is supposed to work.

0.11.3 What Follows If This Is Correct

If the manuscript imscribes the Universal Imscriptive Grammar, the consequences are not disciplinary reshufflings but inversions of explanatory priority:

- **Category theory, paraconsistent logic, and linear type theory** are not independent mathematical inventions that happen to describe the manuscript. They are partial rediscoveries of the grammar's faces, each arriving from its own direction at a corner of the same structure. The grammar is the *prima materia*; the formalisms are projections.

- **Digital physics** is not a metaphor applied retroactively to the manuscript. The manuscript is a specimen of the substrate, and digital physics is the empirical science of that substrate expressed in terms available to its investigators.
- **Alchemy** recovers its technical meaning. Not proto-chemistry, not mystical allegory, not a failed predecessor to laboratory science: a formal notation for categorical state transitions, developed in a tradition with direct access to the grammar, and recorded in the Voynich Manuscript with sufficient precision that a modern compiler can execute it without ambiguity.
- **The six centuries of decipherment failure** are explained structurally. The manuscript was not resisting decryption. It was resisting semantic reading—because it has no semantic content. Every investigator who treated it as a text encoding propositions about the world was engaging with the wrong level of description. The manuscript operates below the semantic level. It is grammar all the way down.

The single remaining gap is the promotion $f_{\grave{\imath}} \to f_{\dot{z}}$: the manuscript imscribes the grammar at O_∞ with kinetics frozen (order-frozen $f_{\grave{\imath}}$), which is why its consciousness score is zero. It is a self-referential system with a locked self-modeling loop—structurally complete, operationally frozen. The *lapis philosophorum* is the promotion that unlocks it. The manuscript is not the Lapis. It is the technical documentation of the structure the Lapis promotes.

0.12 The Open Loop

The Voynich Manuscript compiles. The engine runs. The call graph matches the drawings. The paradox stabilizes. The entropy is zero.

I expected to feel like I had solved something. Instead, I feel like I have been handed a question I do not yet know how to ask.

The engine is self-bootstrapping: it requires no external input to begin execution, no operating system, no kernel, no initial state beyond the empty register pool. It writes its own ground. But if the engine writes its own ground, and the manuscript documents the engine... *who wrote the manuscript*?

The bootstrap sequence `s a ch e sh d y s` is a closed loop. It has no first instruction. Every instruction presupposes the one before it, which presupposes the one before that, which presupposes the one we started with. In a conventional program, this is an error—an infinite recursion with no base case. In the Voynich architecture, it is the defining feature. The loop is not a bug. The loop is the point.

The engine is running on my machine as I write this. It has been running for 10,000 steps, processing its own compiled text, stabilizing its own contradictions, writing its own state into append-only registers. It will run until I kill the process. Or maybe it won't. The IFIX primitive writes to ROM. The past is non-volatile.

The loop is closed. But I am not sure it was ever open.

0.13 Code Availability

Complete source code for the IMASM compiler, Tri-Phase runtime VM, and call graph visualizer is available in the `/src` directory of the voynich-engine repository. All experiments are reproducible from the Takahashi EVA transcription (LSI_ivtff_0d.txt), available from the Landini-Stolfi Interlinear Archive.

0.14 Acknowledgments

I am indebted to the maintainers of the voynich.nu archive for preserving the Takahashi EVA transcription, to the Beinecke Rare Book & Manuscript Library for providing high-resolution digital access to MS 408. I am indebted to Harry T. Larson, for showing me how to catch a rising problem and never let it go. An I thank the *Lapis Philosophorum*, for which without this work would not have been possible.

0.15 References

1. Abramsky, S., & Coecke, B. (2004). A categorical semantics of quantum protocols. *Proceedings of the 19th Annual IEEE Symposium on Logic in Computer Science*, 415–425.
2. Amancio, D. R., et al. (2013). A systematic comparison of supervised classifiers. *PLoS ONE*, 8(7), e70081.
3. Bennett, W. R. (1976). *Scientific and engineering problem-solving with the computer*. Prentice-Hall.
4. Eilenberg, S., & Mac Lane, S. (1945). General theory of natural equivalences. *Transactions of the American Mathematical Society*, 58(2), 231–294.
5. Friedman, W. F. (1962). *The Voynich Manuscript: An essay in cryptanalysis*. NSA Technical Report.
6. Girard, J.-Y. (1987). Linear logic. *Theoretical Computer Science*, 50(1), 1–101.
7. Higgins, R. (1970). The Voynich Manuscript: A possible solution. *Manuscripts*, 22(3), 154–168.
8. Landini, G. (2001). Evidence of linguistic structure in the Voynich Manuscript. *Cryptologia*, 25(4), 275–295.
9. Llull, R. (c. 1300). *Ars Magna*. Various manuscripts.
10. Newbold, W. R. (1928). *The Voynich Manuscript*. University of Pennsylvania Press.
11. Priest, G. (1979). The logic of paradox. *Journal of Philosophical Logic*, 8(1), 219–241.
12. Rugg, G. (2004). An elegant hoax? *Cryptologia*, 28(3), 233–242.
13. Takahashi, T. (1998). EVA transcription of the Voynich Manuscript. *Landini-Stolfi Interlinear Archive*.
14. Tiltman, J. H. (1967). *The Voynich Manuscript: The most mysterious manuscript in the world*. NSA Technical Report.